CivicAI — Simple Project Overview

A smart way to report and fix city problems automatically

What is CivicAI?

CivicAI is a website that lets citizens report municipal issues (like potholes, broken streetlights, or trash piles) and automatically routes them to the right city department. Instead of city workers reading every single report to decide who should fix it, our built-in AI does it instantly.

How to Use CivicAI (The Process)

- **Step 1: Citizen Login**
  A citizen logs into the website securely using their account credentials.
- **Step 2: Submit a Report**
  The citizen fills out a short description of the issue and uploads a photo of it (for example, a photo of a broken streetlight).
- **Step 3: Instant AI Analysis**
  The server saves the report, and the AI immediately scans both the photo and text to identify what and how urgent the problem is.
- **Step 4: Automatic Routing**
  The report is saved in the database, pre-sorted into a department (e.g. *Utilities & Lighting*) and assigned an urgency rating.
- **Step 5: Admin Workstation**
  The city administrator logs into their dashboard. They see all reports neatly sorted on a chart. They can click on the citizen's report, assign it to a repair team, and leave updates.
- **Step 6: Live Tracking**
  The citizen checks their dashboard and sees the progress status (e.g., 'In Progress') and read the admin's remarks in real-time.

Under the Hood (How the AI Works)

To make the system automatic, the backend server uses two simple smart systems to read images and text:

1. Image Object Detection (How it reads photos)
We use a pre-trained computer model called **MobileNetV3**. Think of it as a brain that has looked at millions of photos and already knows what general objects look like. If you upload a photo of a garbage bin, it recognizes it. The backend then maps 'garbage bin' to the *Waste Management & Sanitation* department.

2. Keyword Scanner (How it reads text)
If the photo is blurry, dark, or missing, the system scans the text description for words it knows. For example, if the description contains words like 'leak', 'pipe', or 'drain', it routes the ticket to *Utilities & Lighting*.

3. Urgency Checker (How it determines priority)
The system scans reports for danger words. If it sees high-risk terms like 'fire', 'smoke', 'live wire', or 'explosion', it automatically marks the report as **High** or **Critical**. This pushes the report to the top of the administrator's dashboard so it gets fixed first.

Why this is useful: It removes manual work. Citizens get their problems resolved faster, and city administrators spend less time sorting emails and more time organizing repairs.